

3.3.2 Verlet

Recall the position and velocity Verlet methods:

Position Verlet: **Velocity Verlet:**

$$\begin{aligned}\hat{\mathbf{Q}} &:= \mathbf{q} + (h/2) \mathbf{M}^{-1} \mathbf{p}, & \hat{\mathbf{P}} &:= \mathbf{p} - (h/2) \nabla U(\mathbf{q}), \\ \mathbf{P} &:= \mathbf{p} - h \nabla U(\hat{\mathbf{Q}}), & \mathbf{Q} &:= \mathbf{q} + h \mathbf{M}^{-1} \hat{\mathbf{P}}, \\ \mathbf{Q} &:= \hat{\mathbf{Q}} + (h/2) \mathbf{M}^{-1} \mathbf{P}, & \mathbf{P} &:= \hat{\mathbf{P}} - (h/2) \nabla U(\mathbf{Q}).\end{aligned}\tag{3.3}$$

We can think of the velocity Verlet method as being defined by a splitting into three parts:

$$\begin{aligned}H(\mathbf{q}, \mathbf{p}) &= H_1 + H_2 + H_3, \\ H_1 &= \frac{1}{2} U(\mathbf{q}), \\ H_2 &= \frac{1}{2} \mathbf{p}^T \mathbf{M}^{-1} \mathbf{p}, \\ H_3 &= \frac{1}{2} U(\mathbf{q}).\end{aligned}$$

This is a sequence of two types of operations: it consists of a “kick” exhibited by a jump in the momentum, linear “drift” with the resulting momentum, followed by a final kick. The symmetry of the method is one of its important features. Switching the order of the operations, i.e. drift-kick-drift, gives the position Verlet method.

In general for the Verlet method applied to Hamiltonians of the form $H(\mathbf{q}, \mathbf{p}) = T(\mathbf{p}) + U(\mathbf{q})$, the BCH lemma gives

$$\begin{aligned}\tilde{H}_h &= T + U + \frac{h^2}{12} \left(\{T, \{T, U\}\} - \frac{1}{2} \{U, \{U, T\}\} \right) \\ &\quad + \frac{h^4}{120} \left(-\frac{1}{6} \{T, \{T, \{T, \{T, U\}\}\}\} + \frac{1}{3} \{U, \{T, \{T, \{T, U\}\}\}\} \right. \\ &\quad \left. - \frac{1}{4} \{U, \{U, \{T, \{T, U\}\}\}\} + \{T, \{T, \{U, \{U, T\}\}\}\} \right) + \mathcal{O}(h^6)\end{aligned}$$

The extra power of h in the leading perturbation reflects the fact that this method is 2nd order accurate, rather than first order, as Symplectic Euler. Note that only even order terms ($h^2, h^4, \dots$) appear in the perturbative expansion; this is a consequence of the symmetry of the method.